VIRTUALIZATION SOLUTIONS

A BRIEF INTRODUCTION TO HYPERVISORS AND VIRTUALIZATION

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WHAT IS AN HYPERVISOR

- A hypervisor is a form of virtualization software used in Cloud hosting to divide and allocate the resources on various pieces of hardware.
- The hypervisor is a hardware virtualization technique that allows multiple guest operating systems (OS) to run on a single host system at the same time.
- A hypervisor is sometimes also called a virtual machine manager(VMM).

TYPE-1 HYPERVISOR

The hypervisor runs on the underlying host system. It is known as a “Native Hypervisor” or “Bare metal hypervisor”.

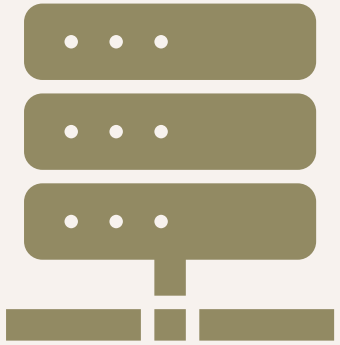
It does not require any base server operating system. It has direct access to hardware resources.

Type 1 hypervisors include VMware ESXi, Citrix XenServer, and Microsoft Hyper-V hypervisor.

Pros: Such kinds of hypervisors are very efficient because they have direct access to the physical hardware resources(like Cpu, Memory, Network, and Physical storage).

Cons: One problem with Type-1 hypervisors is that they usually need a dedicated machine to perform operations and instruct different VMs and control the host hardware resources.

TYPE-2 HYPERVISOR



- A Host operating system runs on the underlying host system. It is also known as 'Hosted Hypervisor".
- Type-2 hypervisors doesn't run directly over the underlying hardware rather they run as an application in a Host system(physical machine).
- Type-2 hypervisor includes VMware Player or Parallels Desktop. Hosted hypervisors are often found on endpoints like PCs
- Pros: Such kind of hypervisors allows quick and easy access to a guest Operating System alongside the host machine running.
- Cons: Here there is no direct access to the physical hardware resources so the efficiency of these hypervisors lags in performance as compared to the type-1 hypervisors

MAP OF VIRTUALIZATION SOLUTIONS

Solution	Type	Pros	Cons	Apple Silicon Compatibility
VMware Workstation Pro	Type-2	Professional-grade features Multi-platform support Advanced networking options	Expensive No native support for Apple Silicon (requires Intel emulation)	NO
VMware Workstation Player	Type-2	Free for personal use Easy-to-use interface Reliable for basic tasks	Limited features compared to Pro version No Apple Silicon support	NO
Oracle VirtualBox	Type-2	Free and open-source Lightweight Cross-platform support	Slower performance Lacks enterprise features	NO (experimental support only)
Microsoft Hyper-V	Type-1	Native to Windows Efficient resource allocation Enterprise-grade performance	Limited to Windows hosts Not cross-platform	NO
Parallels Desktop	Type-2	Optimized for macOS Supports Apple Silicon natively Easy setup	Subscription-based Fewer advanced enterprise features	YES
UTM	Type-2	Free and open-source Native to Apple Silicon Supports ARM and x86 virtualization	Limited enterprise-level features Slower than VMware	YES

VIRTUALIZATION SOLUTIONS CLOSE UP

- VMware Workstation Pro

Type: Type II

Advanced features: snapshots, networking, and **VM cloning.**

Ideal for **enterprise use** and **resource-intensive applications**.

- VMware Workstation Player

Type: Type II

Free for personal use with **basic VM management**

Beginner-friendly and lightweight.

- Oracle VirtualBox

Type: Type-2

Free and open-source with snapshots and **cross-platform** support.

Good for testing **environments**, limited performance..

VIRTUALIZATION SOLUTIONS CLOSE UP

- Microsoft Hyper-V

Type: Type-1

Built into Windows, optimized for **server virtualization.**

Efficient and **enterprise-grade**, **Windows-only**.

- Parallels Desktop

Type: Type-2

Native Apple Silicon support with **seamless macOS integration**.

Best for running Windows on Mac with ease.

- UTM

Type: Type-2

Free and Apple Silicon native, supports **ARM and x86**.

Lightweight and simple, best for **personal projects.**

CONCLUSION AND CHOICE

- **Type I**: Best for enterprise and servers -> Hyper-V.
- **Type II**: Ideal for personal use and development -> VMware, VirtualBox.
- **Mac (Apple Silicon)**: -> Parallels for seamless experience -> UTM for free option.
- **Budget**: VirtualBox and UTM are free, VMware offers advanced features for a cost.
- **Recommendation**: Base your choice on platform, features, and budget.